

JUNJIE LI

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EDUCATION

Northeastern University

Master of Science in Computer Science (GPA: 3.9/4.0)

Teaching Assistant: Object Oriented Design (Spring 2024)

Temple University

M.S. in Business Research (Quantitative Marketing), PhD quit

Nankai University

B.S. in Mathematics (Chern Honor Class)

Seattle, WA | Sep 2023 - Dec 2025

Philadelphia, PA

Tianjin, China

EXPERIENCE

Berkshire Hathaway HomeServices: Capstone Intern

Seattle, WA | Jun 2024 - Aug 2024

Machine Learning Engineer

- Maintained data pipelines using **Apache Airflow**, ensuring seamless data flow.
- Customized and fine-tuned **deep learning models** (N-Beats series) for housing price and inventory predictions, improving model accuracy by 30%. Applied **Agile** principles to iteratively refine data acquisition processes and update models, resulting in a 25% reduction in development cycle time.

PhD Research: Understand User's Video Appreciation using Live Comments

Philadelphia, PA

Analyzed live comments with data mining, computer vision and natural language processing techniques.

- Preprocessed live comment data with **Pandas** and **NumPy**, and video content using **OpenCV** and **BLIP**. Extracted features from frames and text with **ResNet-50** and pre-trained **BERT** embeddings.
- Improved data processing scalability and efficiency by 5x using **Dask** and **NVIDIA RAPIDS** suite.

PROJECTS

Search Engine for Emojis | [Link](#)

LLM-RAG Python/Go Web

"Emoji Search", a RAG-based AI search engine finding emojis based on web services.

- Adopted an event-driven architecture (**EDA**) for smoother search experience with four main components.
- Extended web search feature with Retrieval-Augmented Generation (**RAG**). Integrated **Kafka** to streams user search queries flow, providing real-time contextual information to the generation step in low latency.
- Optimized microservice discovery and balanced queries, resulting in 30% fewer service failures and 15% faster response times at high volumes (5000 QPS). Containerized it in **Docker** and deployed it on **AWS EKS**.

AsyncTask Scheduler | Collaborated with [Hui Zhou](#) | [Link](#)

Distributed Go Framework

"Async4Ai", a light-weighted framework asynchronously processing interdependent tasks.

- Designed distributed task-scheduling system that decouples producers from consumers, highlighting complete task management via web module with **RESTful API** for interaction between agents and database.
- Crafted loosely coupled table normalization & sharding logic in **MySQL** to improve database performance.
- Performed stress testing and improved performance from 500 QPS to 2000 QPS by using distributed locks and caching with **Redis** and reconfiguring the database connection pool.

Anonymous Academic Career Hub | [Link](#)

Full-stack JavaScript Web

"Business Job Market Rumors", a discussion board for academic jobs in business schools.

- Utilized **Express/Node** + **React** in a client-server model, integrating **Tailwind CSS**.
- Implemented role-based access control, context-based authentication using **JWT**, and automated content moderation using **Flask** to ensure a secure and respectful discussion environment.
- Deployed on **Azure App Service** via **GitHub Actions** with **MongoDB** hosted on **Azure Cosmos DB**.

TECHNICAL SKILLS

Languages & Databases: Python(5yr), Java(2yr), Golang(2yr), C/C++(1yr), JavaScript, SQL; MongoDB, Redis

Frameworks: CUDA, Spark, Dask, PyTorch, TensorFlow, Langchain, Streamlit, Express, Node, Angular, FastAPI

Tools: Dataiku, NeMo, Flask, Gradle, JUnit, Postman, GitHub Actions(CI/CD), Docker, Spark, Kubernetes

Miscellaneous: Linux, AWS([Certified Solutions Architect Associate](#)), Azure([Certified AI Engineer Associate](#))